

Def. Let F be a field. Define the General Linear Group as:

$$GL_n(F) \stackrel{\text{def}}{=} \{A \in M_{n \times n}(F) \mid \det A \neq 0\}$$

$I_n \in GL_n(F)$, $\det A, \det B \neq 0 \Rightarrow \det AB \neq 0$, $\det A \neq 0 \Rightarrow \det A^{-1} \neq 0$, exists

Thus it is group.

Def. Define the Special Linear Group as: $SL_n(F) \stackrel{\text{def}}{=} \{A \in M_{n \times n}(F) \mid \det A = 1\}$

Lemma. Let $\mathbb{F}_q$ be a finite field of order q . Then
 $GL_n(\mathbb{F}_q)$ has order $(q^n - 1)(q^n - q) \dots (q^n - q^{n-1})$

$SL_n(\mathbb{F}_q)$ has order $\frac{1}{q-1} |GL_n(\mathbb{F}_q)|$

Proof. Clearly, $|M_{n \times n}(\mathbb{F}_q)| = q^{n^2}$.

Choose the first column as non-zero, there are $q^n - 1$ possibilities.

And, the second column all but a multiple of the first, then $q^n - q$ possibilities.

Successively, choose the third as all but not linear combination of first and second, then there are $q^n - q^2$ (More specifically, q^2 is the number of cases of $a_1 u_1 + a_2 u_2$).

Inductively, we obtain that $|GL_n(\mathbb{F}_q)| = \prod_{k=0}^{n-1} (q^n - q^k)$

Meanwhile, observe that

$$\varphi: GL_n(\mathbb{F}_q) \rightarrow \mathbb{F}_q^\times: A \mapsto \det A$$

is well-defined (because $\det A$ is obtained by operations on $\mathbb{F}_q$), onto ($\det \begin{pmatrix} 1 & & \\ & \ddots & \\ 0 & & a \end{pmatrix} = a$) and Homomorphism ($\det AB = \det A \cdot \det B$) with $\ker \varphi = SL_n(\mathbb{F}_q)$. Since

$$GL_n(\mathbb{F}_q) / \ker \varphi \cong \mathbb{F}_q^\times, \text{ we obtain } |\ker \varphi| = |SL_n(\mathbb{F}_q)| = \frac{1}{q-1} \cdot |GL_n(\mathbb{F}_q)|.$$

Def. The Heisenberg group over F defined by

$$H(F) \stackrel{\text{def}}{=} \left\{ \begin{pmatrix} 1 & a_1 & a_2 \\ 0 & 1 & a_3 \\ 0 & 0 & 1 \end{pmatrix} \mid a_i \in F \right\} \leq SL_3(F).$$

Prop. $Z(H(F)) \cong (F, +)$

Proof observe that

$$\underbrace{\begin{pmatrix} 1 & a_1 & a_2 \\ 0 & 1 & a_3 \\ 0 & 0 & 1 \end{pmatrix}}_A \cdot \underbrace{\begin{pmatrix} 1 & b_1 & b_2 \\ 0 & 1 & b_3 \\ 0 & 0 & 1 \end{pmatrix}}_B = \begin{pmatrix} 1 & a_1+b_1 & a_2+b_2+a_1b_3 \\ 0 & 1 & a_3+b_3 \\ 0 & 0 & 1 \end{pmatrix}$$

Therefore, $\begin{pmatrix} 1 & a_1 & a_2 \\ 0 & 1 & a_3 \\ 0 & 0 & 1 \end{pmatrix}^{-1} = \begin{pmatrix} 1 & -a_1 & -(a_2-a_1a_3) \\ 0 & 1 & -a_3 \\ 0 & 0 & 1 \end{pmatrix}$

And, $AB = BA \Leftrightarrow a_1b_3 = a_3b_1$. This implies

$$A \in Z(H(F)) \Leftrightarrow \forall b_1, b_3 \in F, a_1b_3 = a_3b_1 \\ \Leftrightarrow a_1 = a_3 = 0.$$

Therefore, $Z(H(F)) = \left\{ \begin{pmatrix} 1 & 0 & a \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \mid a \in F \right\}$.

Now, $F \rightarrow Z(H(F)): a \mapsto \begin{pmatrix} 1 & 0 & a \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$ is isomorphism by

$$\begin{pmatrix} 1 & 0 & a \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & b \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 0 & a+b \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

$$GL_3(\mathbb{F}_q) \quad (q^3-1)(q^3-q)(q^3-q^2)$$

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$$SL_3(\mathbb{F}_q) \quad (q^3-1)(q^3-q)q^2$$

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$$H(\mathbb{F}_q) \quad q^3$$

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$$Z(H(\mathbb{F}_q)) \cong (\mathbb{F}_q, +) \quad q$$

|

$$1.$$

Prop. Let $D_{2n} \stackrel{\text{def}}{=} \langle r, s \mid r^n = s^2 = (rs)^2 = 1 \rangle$ be the Dihedral Group.

Define $\varphi: D_{2n} \rightarrow GL_2(\mathbb{R})$ s.t. $\varphi(r) = \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix}$ and $\varphi(s) = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$,

and preserves operation. That is, $\varphi(r^n s^m) := \varphi(r)^n \cdot \varphi(s)^m$.

Then, φ is injective. That is, φ is embedding.

